Crop-ranking reversal under downside risk and operational constraints

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Abstract

Crop-ranking systems order crops by agronomic or predicted performance, whereas acreage decisions depend on uncertain margins, joint downside risk and operational feasibility. We formulate a stochastic crop-planning programme with price, yield and cost uncertainty; land, budget, rotation, contract, labour and equipment constraints; and a loss-CVaR ceiling. In a fixed-land two-crop model, when the higher-ranked crop also has the higher expected margin, reversal occurs if and only if the risk-feasible share interval ends below the equal-share allocation. In a public-data-calibrated Kansas experiment, winter wheat ranks first by a pre-decision historical relative-yield performance index but receives the smallest optimized share, producing a complete rank reversal. The Kansas result is margin-induced and moderated by downside risk. Separate controlled experiments show that tightening a loss-CVaR ceiling can reverse the soybean–corn allocation order while preserving both score and expected-margin order, whereas a soybean rotation cap produces an operationally induced reversal. A selected Gaussian mean–variance policy reduces Gaussian variance by 15.3% relative to the expected-profit benchmark but violates the Student-t evaluation-law loss-CVaR ceiling. Under a shared ex-ante loss-CVaR model, the information–flexibility interaction can be positive, zero or negative, depending on whether flexibility enables reallocation or buffers the predicted state. A 248-state-year public panel documents score–acreage disagreement descriptively, while its lagged association includes zero and does not identify private objectives or constraints. Crop rankings therefore support acreage recommendations only when combined with explicit economic, risk and operational assumptions.

1 Introduction

Crop recommendations are often communicated as an order: crop i is preferred to crop j, or a predictive model assigns i the higher performance score. Acreage planning instead requires a vector of land commitments before prices, yields and operating conditions are fully known. The two orders can differ when margins co-move in adverse years or when budgets, rotations, contracts, labour and machinery restrict feasible crop shares. A producer may therefore plant less of the highest-ranked crop while still using the information contained in the ranking.

We define this event as *crop-ranking reversal.* For $s_i > s_j$, pairwise reversal occurs when $x_i < x_j$ in an optimized acreage plan. A complete rank reversal occurs when the top-ranked crop receives less land than every lower-ranked crop. Strong exclusion reversal additionally requires $x_i = 0 < x_j$. We ask which margin, downside-risk and operational mechanisms move an optimum across these ranking boundaries. We also examine whether variance diversification protects the lower tail and when information becomes more or less valuable as operational flexibility changes.

The analysis keeps the external score separate from the economic objective. The Kansas application uses a pre-decision historical relative-yield performance index based on state yield relative to same-year national yield. Expected margins are constructed independently from yield, national harvest price and operating cost. Joint downside risk is represented separately because marginal dispersion and rank correlation do not determine joint tail exposure. Gaussian, Student-t and Clayton copulas can share Kendall's τ while assigning different probabilities to simultaneous crop losses.

Operational feasibility then links economic margins and joint risk to acreage. The model maximizes expected profit subject to a loss-CVaR ceiling and a polyhedral set containing land, crop bounds, budget, rotation, contract, labour and equipment constraints. Its finite-scenario form is a linear programme. Optimal-face bounds distinguish reversal that is possible at some optimum, universal across all optima or selected by a specified deterministic rule. The complete Karush–Kuhn–Tucker system separates downside-risk effects from simultaneously active operational constraints.

The first theoretical result characterizes reversal in a fixed-land two-crop model. When the higher-ranked crop also has the higher expected margin, its optimal share is the right endpoint of the risk-feasible interval. Reversal occurs exactly when that endpoint lies below the equal-share allocation. Under upper-half risk monotonicity, the criterion reduces to loss-CVaR at equal share exceeding the ceiling. A unique dependence threshold additionally requires continuity, strict monotonicity and crossing within a specified joint-law family. Otherwise, the reversal region can be disconnected.

The Kansas experiment illustrates how the mechanisms differ. In Kansas, winter wheat ranks first by the historical relative-yield performance index but has the lowest calibrated mean margin and receives the smallest optimized land share. The resulting complete rank reversal is margin-induced, while the binding loss-CVaR ceiling increases the winter-wheat share and moderates the reversal. Separate controlled experiments preserve the soybean–corn score and expected-margin order. Tightening the loss-CVaR ceiling produces a risk-induced reversal, whereas tightening a soybean rotation cap produces an operationally induced reversal. Positive crop minimums preclude strong exclusion reversal in the principal specification. Allowing crop exit leaves the strong exclusion count at zero on the evaluated grid.

The diversification analysis compares three policies under common operational constraints. A selected Gaussian mean–variance policy is the first point on the matched-Kendall frontier to meet a fixed 15% variance reduction relative to the expected-profit benchmark. It reduces Gaussian variance but exceeds the Student-t evaluation-law loss-CVaR ceiling. Operationally strong failure occurs for $\gamma \in [0.0068, 0.0088]$. Because the tail-aware policy binds the common ceiling, tail inferiority and ceiling violation measure the same loss-CVaR gap in this experiment.

Information value depends on the action that flexibility enables. Post-signal acreage adjustment can make a more accurate pre-plant signal increasingly valuable. Shock-buffering flexibility can instead reduce the distinction between predicted states and lower the incremental value of information. Under the shared ex-ante loss-CVaR model, signed cross-differences classify these information–flexibility interactions without extending the theorem beyond its separable or nonbinding-risk assumptions.

An external 31-state panel finally assesses descriptive score–acreage disagreement. It documents substantial variation across ranking definitions, but the lagged association between prior score leadership and next-year acreage change includes zero. The panel therefore describes observed frequencies without identifying private objectives, constraints or risk preferences. Figure 1 summarizes the progression from an external score to margins, joint risk, operational feasibility and optimized allocation.

2 Literature review and research position

2.1 Risk-aware crop planning

Crop-planning models have long represented planting as an operational allocation rather than a list of recommended crops. Their variables include crop areas, sequences and timing; their constraints include land, capital, labour, equipment, rotations, storage and contracts. Contemporary models combine these objects with crop-selection integer programmes [Filippi et al., 2017], dynamic rotation benefits [Boyabatlı et al., 2019], sustainable rotation systems [Benini et al., 2023], contract-farming schedules [Li et al., 2015] and spatial decision support [Pahmeyer et al., 2021]. Reviews show that climate uncertainty is handled through stochastic, robust and multiobjective approaches rather than one standardized decision model [Randall et al., 2022, 2024].

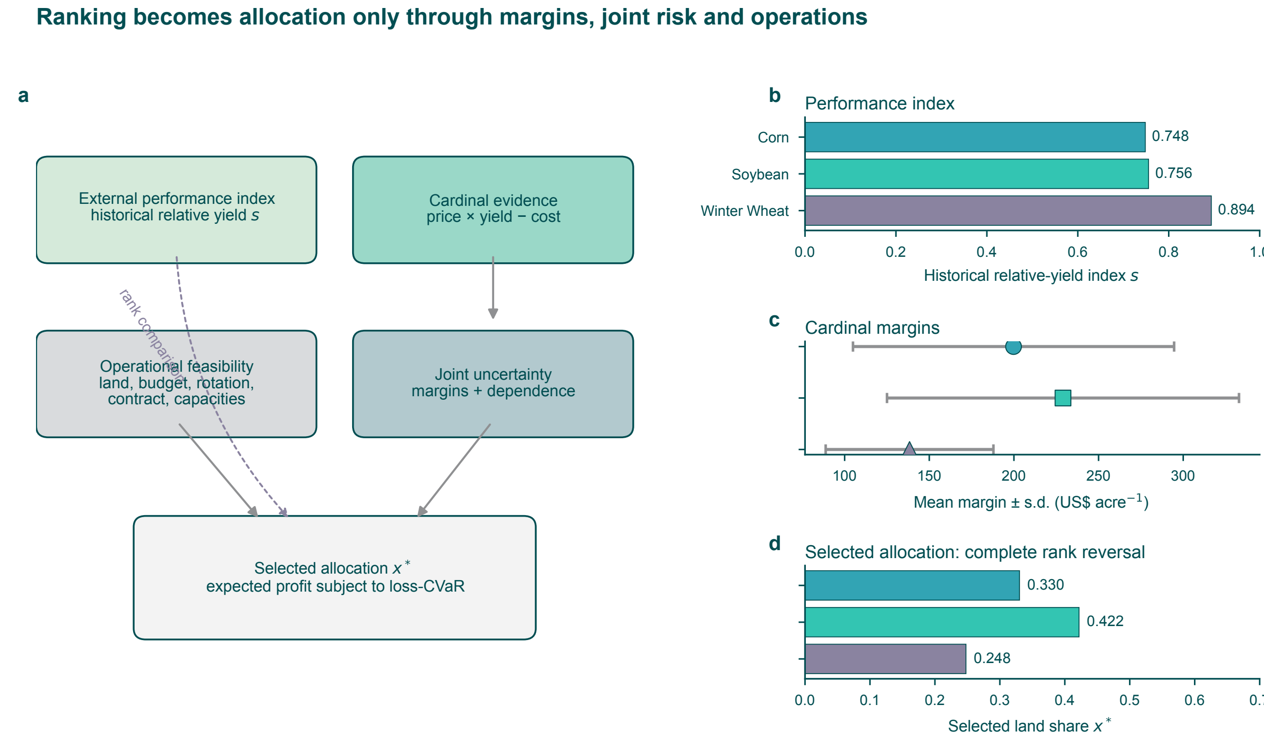

Figure 1: From performance index to risk-constrained operational allocation. **a**, The index remains external to the profit objective. Stochastic margins and their joint law enter expected profit and loss-CVaR, while land, budget, rotation, contract and shared-capacity restrictions define the feasible set. **b**, The historical relative-yield performance index ranks winter wheat first. **c**, Mean margins are calculated separately; bars show one standard deviation across eight calibration years. **d**, The full-model optimum assigns the smallest share to winter wheat, producing a selected complete rank reversal without crop exclusion.

Loss-CVaR is useful in this setting because it controls expected shortfall and admits a linear finite-scenario representation [Rockafellar and Uryasev, 2000, 2002]. Agricultural precedent is substantial but fragmented. AitSahlia et al. [2011] combine ENSO-conditioned planting and hedging with Gaussian-copula scenarios and CVaR. Filippi et al. [2017] embed CVaR in operational crop selection. Larsen et al. [2015] and Nguyen-Huy et al. [2018] use copulas and mean-CVaR to study geographical wheat portfolios. These studies establish that crop portfolios can be optimized under downside risk. The present study extends this perspective by comparing an externally supplied performance ranking with optimized acreage, characterizing reversal boundaries and examining the complete objective-equivalent optimal face.

2.2 Joint tail risk and the limits of conventional diversification

Diversification is effective when crop outcomes do not fail together in the states that matter to the decision. Linear correlation summarizes average co-movement, not the geometry of the joint lower tail. Student-t copulas retain nonzero symmetric tail dependence, while Archimedean families can concentrate dependence asymmetrically [Demarta and McNeil, 2005]. Dependence comparisons are family- and order-specific [Ansari and Rockel, 2024]; a single lower-tail coefficient does not order arbitrary portfolio losses across families.

Agricultural evidence motivates this distinction. Joint yield tails vary with land quality [Du et al., 2018], production shocks are dependent across major breadbaskets [Pflug et al., 2017], and copula estimates show that the risk reduction from crop diversification changes with crop pair, region and drought severity [Schmitt et al., 2024]. These findings do not imply that diversification generally fails. They imply that

Table 1: Positioning relative to selected research streams. • denotes a central object, ◦ denotes conditional or subfield-specific coverage, and −− denotes an object outside the stream's principal scope. The comparison is selective rather than an exhaustive map of the literature.

Stream	Performance rank	Operations	Loss-CVaR	Joint tail law	Reversal frontier	Info.–flex.
Agricultural ML reviews	•	−−	−−	−−	−−	−−
Crop-planning optimization	−−	•	◦	◦	−−	−−
Agricultural copula–CVaR	−−	◦	•	•	−−	−−
Climate forecast valuation	−−	◦	◦	−−	−−	◦
Present study	•	•	•	•	•	•

"number of crops" and variance reduction are incomplete diagnostics for a loss-CVaR-constrained portfolio. The diversification criterion used here requires a conventional policy to pass a fixed variance benchmark yet fail under the Student-t evaluation law and differ from the tail-aware allocation.

2.3 Prediction, prescription and performance scores

Agricultural machine learning commonly predicts yield, classifies land or ranks crop performance using weather, soil, management and remote-sensing features [Liakos et al., 2018, van Klompenburg et al., 2020]. A score can be scientifically useful even when it is not an acreage prescription. The prescriptive-analytics literature instead evaluates how conditional predictions change an optimized decision [Bertsimas and Kallus, 2020, Elmachtoub and Grigas, 2022]. Decision-focused learning extends this principle to end-to-end systems [Mandi et al., 2023].

The present analysis does not develop a learning algorithm. Instead, it separates prediction from prescription. A pre-decision historical relative-yield performance index ranks crop outcomes but does not enter the profit objective. Optimized acreage can therefore be compared with the external ranking without treating predictive accuracy as evidence of prescriptive optimality or requiring the index to contain every economic variable.

2.4 Information value and operational flexibility

Seasonal climate information has value only through decisions it can change. Reviews emphasize that forecast skill, risk preferences, institutions and available management responses jointly determine economic value [Meza et al., 2008]. Bioeconomic applications measure value through changes in sowing and other decisions [An-Vo et al., 2021], while ENSO-based crop-planning models combine forecasts with planting and financial hedges [AitSahlia et al., 2011]. General decision analysis likewise links information value to the portfolio or action set [Keisler, 2004, Merkhofer, 1975]; operational-flexibility theory shows that flexibility value depends on margins, cost and the full uncertainty structure [Van Mieghem, 1998].

These results do not imply unconditional strict complementarity. A flexible action can make a signal actionable, but it can also provide a robust fallback that reduces the need to distinguish states. We therefore prove non-negativity, strictness and zero conditions separately, add an increasing-differences condition for complementarity, and construct an agricultural buffering path for substitution.

2.5 Contribution matrix

Table 1 identifies how the study combines established research strands; it does not imply exhaustive novelty. The model connects a historical relative-yield performance index to stochastic operational allocation, characterizes a reversal frontier, distinguishes variance and downside-risk criteria for diversification, and identifies conditions under which information complements or substitutes for flexibility. The next section formalizes these elements in a common crop-planning model.

3 Stochastic crop-planning model

3.1 Scores, margins and joint uncertainty

Let $i = 1, \dots, n$ index crops and x_i denote planted area. A pre-decision historical relative-yield performance index s_i induces a crop order. The index is external to the optimization model; no cardinal transformation of s is used as a margin. For state ω, stochastic per-acre margin and portfolio profit are

$$\widetilde{\pi}_i(\omega) = \widetilde{P}_i(\omega)\widetilde{Y}_i(\omega) - \widetilde{C}_i(\omega), \qquad \widetilde{\Pi}(x, \omega) = \sum_i x_i \widetilde{\pi}_i(\omega). \tag{1}$$

The joint distribution includes within-crop price–yield dependence and cross-crop co-movement. Let $\mu_i = \mathbb{E}[\widetilde{\pi}_i]$ and define loss $L(x, \omega) = -\widetilde{\Pi}(x, \omega)$. A larger loss-CVaR is worse; because profitable portfolios can have negative loss-CVaR, "more negative" means safer under this convention.

3.2 Operational feasibility

The operational set is

$$\mathcal{X} = \left\{ x \in \mathbb{R}^n_+ : \mathbf{1}^\top x \le A, \quad c^\top x \le B, \quad Rx \le r, \quad Kx \ge k, \quad Gx \le h, \quad \ell \le x \le u \right\}. \tag{2}$$

Here A is available land, c and B describe pre-plant cash requirements, R contains rotation caps, K contains contract minimums, and G contains shared labour, equipment, irrigation or storage rows. Idle land is $A - \mathbf{1}^\top x$ and is reported even when it is not assigned a separate variable. Together, the constraints define the feasible planting decisions before acreage is optimized.

3.3 Heuristic-policy projection

Let x^h be a raw score-proportional, score-winner or equal-share prescription. Its principal comparison policy is the unique Euclidean projection

$$\widehat{x}_2^h = \arg\min_{x \in \mathcal{X}_h} \frac{1}{2} \|x - x^h\|_2^2, \tag{3}$$

where $\mathcal{X}_h$ imposes $\mathbf{1}^\top x = A$, the crop bounds, budget, rotation, contract, labour and equipment constraints in Eq. 2. Idle land is therefore excluded from the comparison set. The loss-CVaR ceiling is not part of the projection; loss-CVaR and risk feasibility are evaluated after projection. Strict convexity gives a unique principal projection. An unweighted L^1 projection provides a sensitivity comparison; on a non-unique L^1-optimal face, a fixed crop-order weighted linear objective selects one allocation deterministically.

3.4 Loss-CVaR ceiling

For confidence level α,

$$r_\alpha(x) = \mathrm{CVaR}_\alpha[L(x)] = \min_{v \in \mathbb{R}} \left\{ v + \frac{1}{1-\alpha} \mathbb{E}[(L(x) - v)_+] \right\}. \tag{4}$$

The producer solves

$$\max_{x \in \mathcal{X}} \ \mu^\top x \quad \text{subject to} \quad r_\alpha(x) \le \kappa. \tag{5}$$

For scenarios $m = 1, \dots, M$ with probabilities w_m, free threshold v and excess variables $q_m \ge 0$, the risk restriction is

$$v + \frac{1}{1-\alpha} \sum_m w_m q_m \le \kappa, \tag{6}$$

$$q_m \ge -\pi_m^\top x - v. \tag{7}$$

The resulting programme is linear. The risk ceiling is indexed by $\rho \in [0, 1]$:

$$\kappa(\rho) = \kappa_{\min} + \rho\left(\kappa_{\mathrm{EP}} - \kappa_{\min}\right), \tag{8}$$

where $\kappa_{\min}$ is minimum attainable loss-CVaR and κ_{EP} is the loss-CVaR of the unconstrained expected-profit endpoint. This makes risk tolerance comparable across joint laws without transferring an arbitrary dollar ceiling.

3.5 Reversal and optimal faces

Definition 1 (Pairwise, complete rank and strong exclusion reversal)**.** *For a pair $s_i > s_j$, an allocation reverses the ranking if $x_i < x_j$. A complete rank reversal occurs when the highest-ranked crop has less area than every lower-ranked crop. A strong exclusion reversal requires $s_i > s_j$ and $x_i = 0 < x_j$.*

Positive crop lower bounds preclude strong exclusion reversal in the principal specification. Sensitivity analyses evaluate this outcome after setting either all lower bounds or only the highest-ranked crop's lower bound to zero.

Let

$$\mathcal{S}_\kappa = \arg\max\{\mu^\top x : x \in \mathcal{X},\ r_\alpha(x) \le \kappa\}$$

be the complete optimal set. Reversal is *possible* if it occurs at some $x \in \mathcal{S}_\kappa$, *universal* if it occurs at every such point, and *selected* if it occurs at the point chosen by a specified deterministic selector. Pairwise classification follows the minimum and maximum of $x_j - x_i$ on the objective-equivalent face; one optimal vertex does not establish universal reversal. Ordering and operational zero use numerical tolerances of 10^{-4} normalized land share; sensitivity spans 10^{-8} to 10^{-2}.

3.6 Full stationarity system

Let nonnegative multipliers $\eta, \nu^A, \nu^B, \nu^R, \nu^K, \nu^G, \nu^\ell, \nu^u$ correspond to loss-CVaR, land, budget, rotation, sign-converted contract, shared capacity, lower bounds and upper bounds. For a loss-CVaR subgradient $d \in \partial r_\alpha(x^*)$, acreage stationarity is

$$-\mu + \eta d + \nu^A \mathbf{1} + \nu^B c + R^\top \nu^R - K^\top \nu^K + G^\top \nu^G - \nu^\ell + \nu^u = 0. \tag{9}$$

Pairwise subtraction cancels the common land multiplier but retains budget, shared-resource and crop-bound pressures. Equation 9 is a marginal balance rather than an acreage-ordering formula. The finite-scenario KKT system also includes stationarity for v and every q_m, including fractional tail weights when the loss distribution has atoms. These conditions provide the optimality basis for the reversal frontier developed next.

4 Theoretical results

4.1 An exact two-crop reversal frontier

Consider two crops with $s_1 > s_2$, fixed planted land $x_1 + x_2 = A$, and an operationally feasible interval $I = [\underline{z}, \overline{z}]$ for $z = x_1$. Under a declared joint-law path θ, write

$$r_\theta(z) = \mathrm{CVaR}_\alpha[-z\widetilde{\pi}_1 - (A - z)\widetilde{\pi}_2], \qquad \mathcal{K}_{\theta,\kappa} = \{z \in I : r_\theta(z) \le \kappa\}.$$

Theorem 1 (Necessary-and-sufficient restricted reversal)**.** *Suppose $\mu_1 > \mu_2$ and $\mathcal{K}_{\theta,\kappa}$ is nonempty. Then the unique optimum is*

$$z^*(\theta, \kappa) = \sup \mathcal{K}_{\theta,\kappa}.$$

Consequently,

$$x_1^* < x_2^* \Longleftrightarrow \sup \mathcal{K}_{\theta,\kappa} < A/2 \Longleftrightarrow \mathcal{K}_{\theta,\kappa} \cap [A/2, \overline{z}] = \varnothing.$$

If r_θ is nondecreasing on $[A/2, \overline{z}]$ and a risk-feasible point exists below $A/2$, reversal is equivalent to $r_\theta(A/2) > \kappa$.

Proof. The objective is $\mu_2 A + (\mu_1 - \mu_2)z$, which is strictly increasing in z. Loss-CVaR is convex and continuous in z, so its sublevel set intersected with compact I is compact. The strictly increasing objective is uniquely maximized at the right endpoint. The three reversal statements then follow from $x_1 < x_2 \Longleftrightarrow z < A/2$. Under upper-half monotonicity, the lowest risk on that half is attained at $A/2$, giving the last equivalence. □

The theorem applies to a fixed-land two-crop model with ordered expected margins, operational bounds and a nonempty risk-feasible set. If $r_\theta(A/2)$ is continuous and strictly increasing along a specified dependence path and crosses κ, there is a unique family-specific threshold θ^* solving $r_{\theta^*}(A/2) = \kappa$. Without strict monotonicity or crossing, the reversal set is $\{\theta : \sup \mathcal{K}_{\theta,\kappa} < A/2\}$; it may contain several components. Higher tail dependence therefore need not produce a monotone or universal reversal response.

4.2 Full-model optimality and selection

Theorem 2 (Complete convex KKT certificate)**.** *Under a convex constraint qualification, or for the bounded finite-scenario linear programme, a feasible solution of Eq. 5 is optimal if and only if there exist nonnegative multipliers for every inequality, together with the free-threshold and scenario-excess dual variables, that satisfy primal feasibility, dual feasibility, stationarity Eq. 9 and complementarity.*

The proof follows convex KKT sufficiency or finite-dimensional linear programming strong duality. Incomplete stationarity can assign a residual margin difference to "risk" even when budget, contract or shared capacity is active. KKT provides pointwise optimality conditions; optimal-face analysis determines whether the ranking relation holds throughout all equivalent optima.

4.3 Dependence paths

Proposition 1 (Named-family feasible-set ordering)**.** *Fix all crop marginals. If $L_{\theta_1}(x) \preceq_{\mathrm{cx}} L_{\theta_2}(x)$ for every x in a declared domain, then*

$$\mathrm{CVaR}_{\alpha,\theta_1}[L(x)] \leq \mathrm{CVaR}_{\alpha,\theta_2}[L(x)].$$

If the domain is all of $\mathcal{X}$, the risk-feasible set under θ_2 is contained in that under θ_1, so optimal expected profit cannot increase. Individual crop shares need not be monotone.

Loss-CVaR is monotone in convex order, yielding the risk and feasible-set inclusions. But set inclusion does not order coordinates of different optimizers. We therefore report Gaussian, Student-t and Clayton results family by family, including active-set transitions, multiple optima and disconnected regions.

4.4 An executable diversification-failure criterion

Let x^{MV} denote the *selected Gaussian mean–variance policy*: the first point on the specified Gaussian frontier that attains a 15% variance-reduction target relative to the expected-profit benchmark x^0. Let x^T denote the tail-aware policy under the Student-t evaluation law.

Definition 2 (Diversification failure)**.** *Conventional diversification fails when*

$$\mathrm{Var}_G[\Pi(x^{MV})] < \mathrm{Var}_G[\Pi(x^0)], \tag{10}$$

$$\mathrm{CVaR}_{\alpha,T}[L(x^{MV})] > \mathrm{CVaR}_{\alpha,T}[L(x^T)], \tag{11}$$

$$x^{MV} \neq x^T, \tag{12}$$

where x^0 is the expected-profit policy under the matched Gaussian law. In computation, allocation disagreement requires $\|x^{MV} - x^T\|_1 > \varepsilon_x$, and tail inferiority requires $\mathrm{CVaR}_{\alpha,T}[L(x^{MV})] > \mathrm{CVaR}_{\alpha,T}[L(x^T)] + \varepsilon_r$. Failure is operationally strong when $\mathrm{CVaR}_{\alpha,T}[L(x^{MV})] > \kappa + \varepsilon_r$.

Proposition 2. *If x^T is feasible and the preceding three conditions hold, x^{MV} is not tail-risk optimal. Under the strong inequality it also violates the declared risk ceiling.*

The criterion separates Gaussian variance reduction, allocation disagreement and downside-risk infeasibility under the heavy-tail evaluation law. It applies to the specified benchmark comparison rather than every diversified portfolio. If $\mathrm{CVaR}_{\alpha,T}[L(x^T)] = \kappa$, the tail-inferiority and ceiling-violation inequalities coincide numerically and must not be counted as independent evidence.

4.5 Information, complementarity and substitution

A state ω is realized and produces a noisy signal S, while a fraction of the planting plan remains adjustable. The producer observes S, chooses a signal-contingent action and subsequently realizes margin. Loss-CVaR is evaluated ex ante over the combined state–signal distribution.

Theorem 3 (Restricted information value)**.** *In a posterior-separable model, or when the shared ex-ante risk restriction is nonbinding, if ignoring the signal is admissible, optimized information value is nonnegative. It is zero when one common action is optimal for every posterior. If posterior optima are unique and differ on positive-probability signal events, with strict improvement on at least one event, information value is strictly positive.*

The constant policy reproduces the no-information solution, proving nonnegativity. A common posterior optimum proves zero value. Strict posterior improvement integrates to positive value in the stated submodel. It does not establish strictness when signal-contingent actions share a binding ex-ante loss-CVaR budget. Nested action sets raise both informed and uninformed optimized values, but their difference need not be monotone.

Proposition 3 (Restricted conditional complementarity)**.** *In the posterior-separable or nonbinding-risk submodel, if actions are nested and state–action payoff has increasing differences in signal precision and acreage flexibility, and posterior-optimal actions differ on events of positive probability, information and flexibility are strictly complementary.*

Proposition 4 (Substitution through shock buffering)**.** *Let flexibility ϕ contract state-dependent margins toward a common vector,*

$$\pi_\omega(\phi) = (1-\phi)\pi_\omega + \phi\bar{\pi}.$$

At $\phi = 1$, the state no longer changes margin and information has zero value. If information value is positive at $\phi = 0$ and the value function is continuous, it must decrease on at least one interval; shock-buffering flexibility substitutes for information there.

These results do not support unconditional supermodularity. For the complete shared ex-ante loss-CVaR numerical programme we instead compute, on every adjacent grid rectangle,

$$\Delta_{\xi,\phi}V = [V(\xi_2,\phi_2) - V(\xi_1,\phi_2)] - [V(\xi_2,\phi_1) - V(\xi_1,\phi_1)].$$

Positive, zero/boundary and negative cross-differences are numerical classifications, not an extension of the restricted theorem. The numerical experiments next evaluate the reversal, diversification and information–flexibility results under a common calibration.

5 Public-data calibration and numerical experiments

5.1 Calibration

Kansas was selected because corn, soybean and winter wheat have complete public observations in every 2016–2024 analysis year and form a relevant diversified crop set. Observations from 2016–2023 calibrate the performance index and margins; 2024 is reserved for external evaluation.

The historical relative-yield performance index for crop i is the mean of Kansas yield divided by same-year national NASS yield over 2016–2023. It measures pre-decision relative-yield performance rather than revenue. Index values are 0.748 for corn, 0.756 for soybean and 0.894 for winter wheat. Real per-acre margin combines Kansas yield, national harvest price, national operating cost and CPI-U conversion to 2024 dollars. Mean margins are \$ 199.8, \$ 229.0 and \$ 138.3, respectively. These aggregate inputs are transparent calibration values, not farm-level profits.

The primary marginal law is Student-t with five degrees of freedom. The primary joint law is a Student-t copula with four degrees of freedom and Kendall's $\tau = 0.25$, simulated with 512 optimization scenarios. Sensitivity uses Gaussian and empirical-quantile margins; Gaussian, Student-t and Clayton copulas; and $\tau \in \{0, 0.15, 0.30, 0.45, 0.60\}$. The eight annual observations cannot estimate extreme tails precisely, so these are structural stress paths rather than fitted farm copulas.

Land is normalized to one. Crop lower bounds are 0.05, 0.10 and 0.05; upper bounds are 0.55, 0.65 and 0.60. The operating budget is \$330, corn is capped at 0.50 by rotation, soybean has a 0.10 contract minimum, and planting labour and harvest equipment have shared crop-specific coefficients. The primary confidence level is 0.95 and risk tolerance $\rho = 0.5$. All counterfactuals use the same full constraint system unless a constraint is the sensitivity dimension.

5.2 Primary reversal

The expected-profit endpoint without loss-CVaR assigns 0.280 to corn, 0.650 to soybean and 0.070 to winter wheat. It produces a complete rank reversal, but not a strong exclusion reversal because winter wheat remains positively allocated; its loss-CVaR $\$ -34.5$, exceeding the primary risk ceiling. The minimum loss-CVaR endpoint attains $\$ -59.5$ but sacrifices expected profit. The midpoint ceiling is $\$ -47.0$.

The full optimum assigns 0.330 to corn, 0.422 to soybean and 0.248 to winter wheat. The highest-ranked crop therefore receives the least land, producing a selected complete rank reversal involving both pairs but no strong exclusion reversal. The land and loss-CVaR restrictions bind; idle land is zero. Expected profit is \$ 198.5 per normalized acre and realized finite-scenario loss-CVaR equals the ceiling to numerical precision. Maximum primal, stationarity and complementarity residuals are $1.42e-14$, $2.84e-14$ and $5.06e-14$, respectively.

This Kansas result is margin-induced: the score–margin separation favours soybean economically even though winter wheat ranks agronomically first, and the expected-profit endpoint is already inverted. The loss-CVaR ceiling raises winter wheat from 0.070 to 0.248 and therefore moderates rather than creates the complete rank reversal. Operational constraints delimit the available path between those endpoints.

5.3 Conditional phase diagram

Figure 2 reports all 165 combinations of copula family, Kendall's τ and risk tolerance. Selected pairwise reversal occurs in 143 cells, selected complete rank reversal in 96 cells. Strong exclusion is not an admissible event in this principal specification because the lower bounds 0.05, 0.10 and 0.05 prevent crop exit; its zero count is therefore structural. There are no infeasible cells, but two Clayton cells have multiple optima. Most reversal regions are connected on the evaluated grid; Clayton at $\tau = 0.30$ is disconnected. The first selected-reversal index ranges from 0 to 0.3 across family and dependence paths.

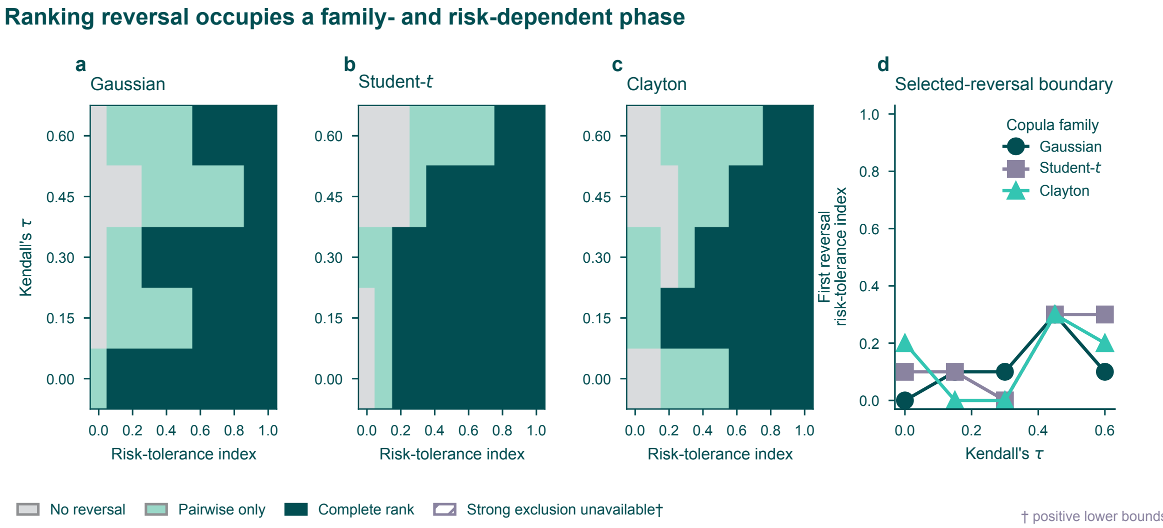

Figure 2: Ranking reversal occupies a family- and risk-dependent phase. a–c, Classification across risk tolerance and Kendall's τ for Gaussian, Student-t and Clayton copulas; the compact key distinguishes no reversal, pairwise-only reversal and complete rank reversal. Strong exclusion is structurally inadmissible here because all crop minima are positive; zero-lower-bound results are reported separately. **d**, The first selected-reversal risk tolerance varies non-monotonically across the named family paths. All 165 parameter combinations are shown, including 2 multiple-optimum cases.

Setting every crop lower bound to zero makes strong exclusion feasible without altering the score, scenario laws, risk path or remaining operational constraints. On that grid, 143 cells show selected pairwise reversal, 95 show selected complete rank reversal, and 0 show selected strong exclusion reversal; possible and universal strong exclusion counts are 0 and 0. The smallest selected share of top-ranked winter wheat is 0.070, so no exclusion boundary or excluded crop exists on the evaluated grid. The targeted sensitivity that removes only winter wheat's lower bound similarly gives 143 pairwise, 96 complete rank reversal and 0 strong exclusion reversal cells. These are substantive null results under specifications in which exit is admissible.

The nonmonotonic cells are substantively important. Increasing dependence can change which resource or risk constraint binds, and that active-set transition can move individual crop shares in either direction. The phase diagram therefore supports Proposition 1's feasible-set logic without claiming coordinate monotonicity. Scalar lower-tail dependence is reported as a diagnostic within each family, not used as a cross-family ordering statistic.

5.4 Diversification failure under matched rank dependence

The comparison x^0 is the matched-Gaussian expected-profit allocation. The selected Gaussian mean–variance policy and tail-aware model use the same marginal means and standard deviations and matched Kendall's $\tau = 0.25$. The Student-t law nevertheless has lower-tail coefficient 0.195, whereas the Gaussian copula has zero asymptotic tail dependence. We solve a complete 301-point Gaussian mean–variance frontier over $\gamma \in [0, 0.03]$, with increments of 10^{-4}. The selected $\gamma = 0.0082$ is the smallest value attaining the fixed, outcome-independent target of at least 15% variance reduction relative to x^0.

At the selected point, Gaussian variance falls from 6765.6 to 5733.5 (15.3%). The canonical policy allocates 0.314 to corn, 0.543 to soybean and 0.143 to winter wheat; Gaussian expected profit is $ 206.6 per acre. The selected policy satisfies all operational constraints but violates the Student-t loss-CVaR ceiling. Its evaluation loss-CVaR is US$ -43.4 per acre, exceeding the ceiling of US$ -45.7 per acre by US$ 2.364 per acre. The allocation changes by 0.090 in L^1 distance. Under the Student-t evaluation law, expected

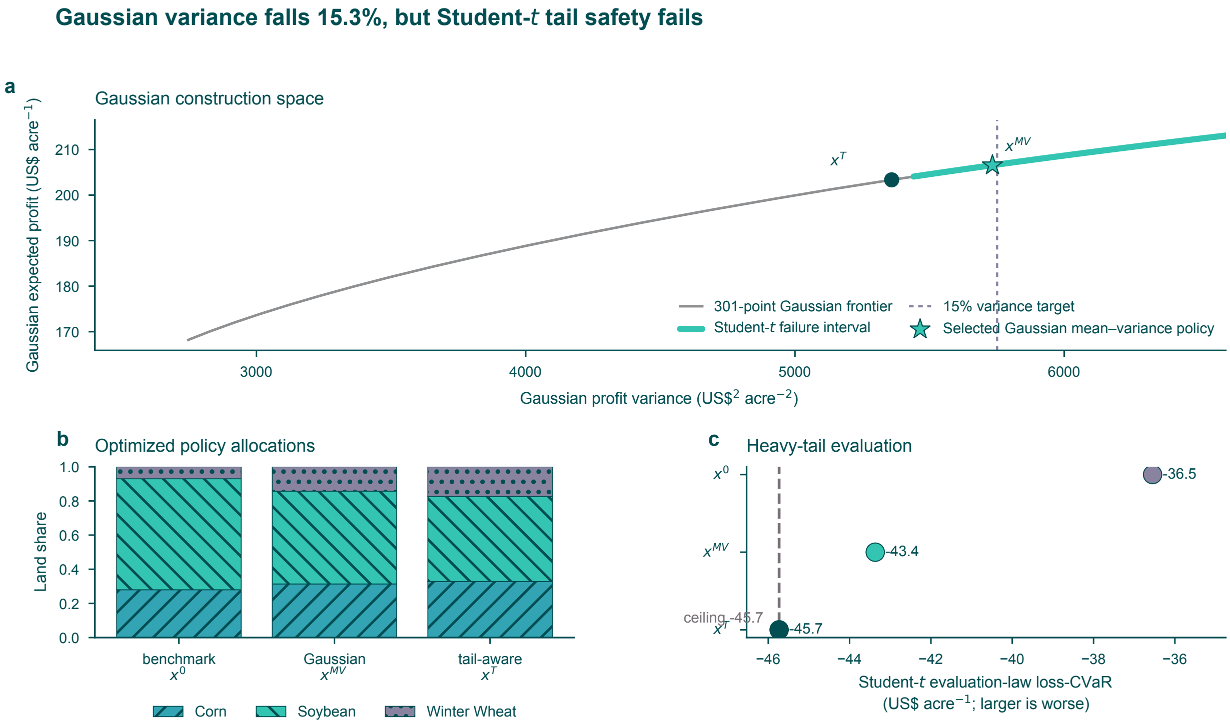

Figure 3: Variance diversification can fail in the joint lower tail. **a**, Complete Gaussian mean–variance frontier, the fixed variance reduction target and the resulting failure interval. **b**, Optimized allocations x^0, x^{MV} and x^T. **c**, Student-t evaluation loss-CVaR for the same policies under a common ceiling. The tail-inferiority and ceiling-violation gaps coincide numerically because x^T binds that ceiling. Gaussian and Student-t laws match Kendall's τ, but only the latter has positive asymptotic lower-tail dependence.

profit is \$ 206.6. The tail-aware policy changes the allocation and lowers evaluation loss-CVaR by \$ 2.364. Both the weak and operationally strong diversification-failure criteria hold for $\gamma \in [0.0068, 0.0088]$. The two classifications share the same interval in this experiment because the tail-aware optimum binds the common loss-CVaR ceiling. Its tail improvement and the Gaussian policy's ceiling violation are numerically equal at the selected point and therefore express the same tail-risk gap. Across 18 one-factor perturbations of target variance reduction, scenario count, seed, dependence, confidence level, risk tolerance and marginal family, 17 satisfy the strong criterion. The result identifies a specified region in which Gaussian variance diversification does not protect the joint lower tail. Its scope is the specified policy comparison and sensitivity domain.

5.5 Mechanism isolation and policy robustness

The score-proportional, winner-take-all and equal-share policies are projected by Eq. 3 before evaluation. The score-proportional policy earns \$ 187.4 with loss-CVaR $\$ -53.1$ and does not reverse. The raw winner-take-all prescription $(0, 0, 1)$ projects to $(0.200, 0.200, 0.600)$ under the principal Euclidean rule, earns \$ 169.6, has loss-CVaR $\$ -59.4$, and does not reverse. Equal share earns \$ 190.4 with loss-CVaR $\$ -51.4$. The expected-profit endpoint earns \$ 216.5 but violates the loss-CVaR ceiling. The risk restriction lowers expected profit relative to the unconstrained expected-profit endpoint.

Projection choice matters for one benchmark. The alternative L^1 rule maps the same winner target to $(0.300, 0.100, 0.600)$, which reverses the soybean–corn pair; both projections remain risk-feasible. Score-proportional and equal-share policies are already feasible and are unchanged. Heuristic reversal is therefore reported as projection-sensitive rather than as a property of the raw score-winner rule.

The controlled risk experiment uses the Kansas soybean–corn index and margin ordering: soybean has both the higher score and a mean-margin advantage of \$ 29.2. A mean-preserving 10% severe soybean drought-or-basis event shifts \$ 154.7 per acre into the downside state and compensates non-adverse states exactly. At loose risk tolerance, soybean and corn receive 0.650 and 0.280; under the tight ceiling they receive 0.181 and 0.219. Tightening the loss-CVaR ceiling therefore reverses the soybean–corn allocation order while preserving the index and expected-margin order. A probability-by-magnitude sensitivity map evaluates 49 stress combinations. Of these, 24 produce an allocation crossing, 24 remain feasible without a crossing, and one is infeasible. In the focal cell, the first crossing occurs at risk tolerance 0.225. These controlled stresses isolate a mechanism; they do not estimate the probability or severity of a Kansas farm shock.

The operational experiment holds the score, margin scenarios and a deliberately nonbinding risk ceiling fixed. Adding the baseline calibrated constraint sequence changes active sets but does not reverse the soybean–corn pair. A controlled soybean rotation-cap path crosses the ranking boundary first at cap 0.35. The preceding stages remain non-reversing; the path demonstrates a structural mechanism rather than estimating a private Kansas constraint (Fig. 4).

Selected pairwise reversal persists across every one-at-a-time sensitivity dimension: α, constraint regime, dependence family, dependence parameter, marginal family, sample window, scenario count, seed and solver. Strong exclusion is structurally inadmissible in those positive-lower-bound runs. The all-zero and top-crop-zero analyses make exclusion admissible, but no strong exclusion reversal occurs across numerical-zero tolerances from 10^{-8} to 10^{-2}. The solver comparisons use HiGHS, dual simplex and interior point; scenario counts are 256, 512 and 1,024; sample windows begin in 2016, 2017 or 2018. These robustness runs assess structural stability, not sampling-independent replication.

5.6 Information and flexibility

The state proxy is whether the annual Kansas corn-minus-soybean relative-yield advantage lies below or above its 2016–2023 median. A noisy pre-plant signal has accuracy 0.5–1.0. The two-stage loss-CVaR restriction is imposed once on the ex-ante distribution across state, signal and contingent actions; it is not replaced by a separate risk budget for each signal.

With post-signal acreage reallocation, no adjustable share or an uninformative signal gives zero value. As accuracy and the adjustable share rise, posterior allocations diverge and information value reaches \$ 4.655 per acre. With shock buffering, more flexibility instead contracts state-specific margins toward a common vector. At perfect signal accuracy, information value declines from approximately \$4.4 without buffering to numerical zero under full buffering. The adjacent grid contains 32 positive, 17 negative, 18 zero-information and 5 zero/boundary classifications. These are discrete cross-difference results for the coupled model, not applications of the restricted complementarity theorem.

5.7 Uncertainty

Sixty-four historical resamples re-estimate scores and margins, regenerate scenarios and resolve the full optimum. Selected pairwise reversal occurs in 62 of 64 resamples, a frequency of 0.969. The exact Clopper–Pearson 95% binomial interval for that resample frequency is $[0.892, 0.996]$. This binomial interval describes the frequency among historical resamples rather than a farm population. The first-reversal risk-tolerance index averages 0.066 with interval $[0.000, 0.598]$, revealing substantial uncertainty in the boundary even though reversal itself is common. Mean bootstrap allocations are 0.277 for corn, 0.527 for soybean and 0.196 for winter wheat; their percentile intervals are wide because only eight years enter each calibration. The eight-year calibration therefore yields stable qualitative reversal but imprecise crop shares and reversal boundaries. The external panel next examines score–acreage disagreement beyond the Kansas calibration without assigning a causal mechanism to observed acreage.

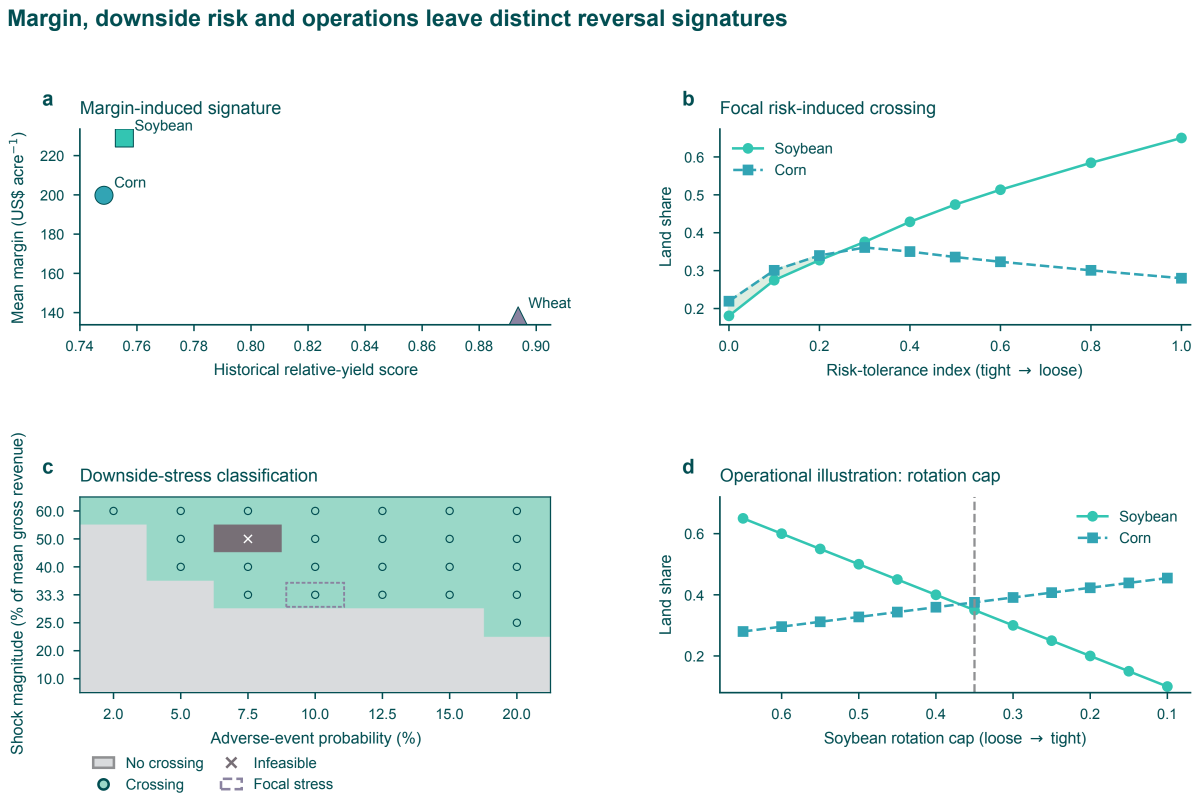

Figure 4: Three reversal mechanisms have distinct counterfactual signatures. **a**, Kansas score and mean-margin ranks separate, so the primary complete rank reversal is margin-induced. **b**, With soybean higher in both score and mean margin, tightening the shared loss-CVaR ceiling under a mean-preserving downside stress crosses the soybean–corn allocation order. **c**, Probability-by-magnitude sensitivity distinguishes crossing, non-crossing and infeasible stress cells. **d**, With risk nonbinding, tightening only the soybean rotation cap produces an operational crossing; the preceding constraint stages do not cross the allocation order.

6 External descriptive evidence

6.1 Purpose and construction

The external analysis asks whether score–acreage disagreement is visible beyond the Kansas calibration. It does not treat observed state acreage as a solution to Eq. 5. USDA NASS annual state reports yield 248 complete state-years across 31 states from 2016 through 2024 for corn, soybean and winter wheat. National harvest prices and operating and total costs come from USDA ERS, and BLS CPI-U supplies the real-dollar conversion. Suppressed or missing acreage and yield are not imputed.

Four ranking definitions are computed before comparison with planted-acreage share. Relative yield is state yield divided by same-year national yield. Standardized revenue multiplies state yield by national price. Operating margin subtracts national operating cost; total-cost margin subtracts total cost. The latter three are transparent accounting measures rather than local or private expected profits. For each state-year we calculate pairwise inversion intensity and whether the top-ranked crop differs from the acreage leader. State-cluster bootstrap intervals preserve within-state serial dependence.

Figure 5: Information–flexibility interaction is classified on the solved value grid. a,b, Information value under post-signal acreage reallocation and shock buffering. **c,d**, Adjacent-grid cross-differences for the same paths, computed as $[V(\xi_2, \phi_2) - V(\xi_1, \phi_2)] - [V(\xi_2, \phi_1) - V(\xi_1, \phi_1)]$. Values are optimized under one shared ex-ante loss-CVaR ceiling.

6.2 Descriptive frequency and definition sensitivity

Top-rank disagreement is 0.887 for relative yield, 0.613 for standardized revenue, 0.625 for operating margin and 0.782 for total-cost margin. The corresponding state-cluster intervals are all well above zero. The ordering across definitions is itself a result: a claim about ranking reversal inherits the scientific meaning of the score being ranked. Relative-yield disagreement cannot be interpreted as profit inefficiency, while a margin-based disagreement remains incomplete because local basis, heterogeneous costs, contracts, beliefs and risk limits are unobserved.

The leakage-free temporal check uses the prior relative-yield index to predict the following change in acreage share with crop, state and decision-year fixed effects. It includes 651 crop-transition rows. The estimated effect is -0.0042 share units with interval $[-0.0123, 0.0056]$, which includes zero. Thus the panel does not show that a crop's earlier index leadership leads to a larger next-year share change. This null is consistent with many mechanisms, including persistent rotations, adjustment cost and an index that does not measure private profit; it is not evidence that agronomic information has no value.

6.3 Evidence boundary

The state panel documents score–acreage disagreement and its sensitivity to the ranking definition. It also yields no detectable leakage-free lagged association between prior relative-yield leadership and the following change in acreage share. Aggregate state data do not identify the model's loss-CVaR mechanism, private risk preferences, active constraints or welfare effects. Identification of those quantities requires farm-level expectations, resource capacities, contract terms and decision timing. The evidence is therefore

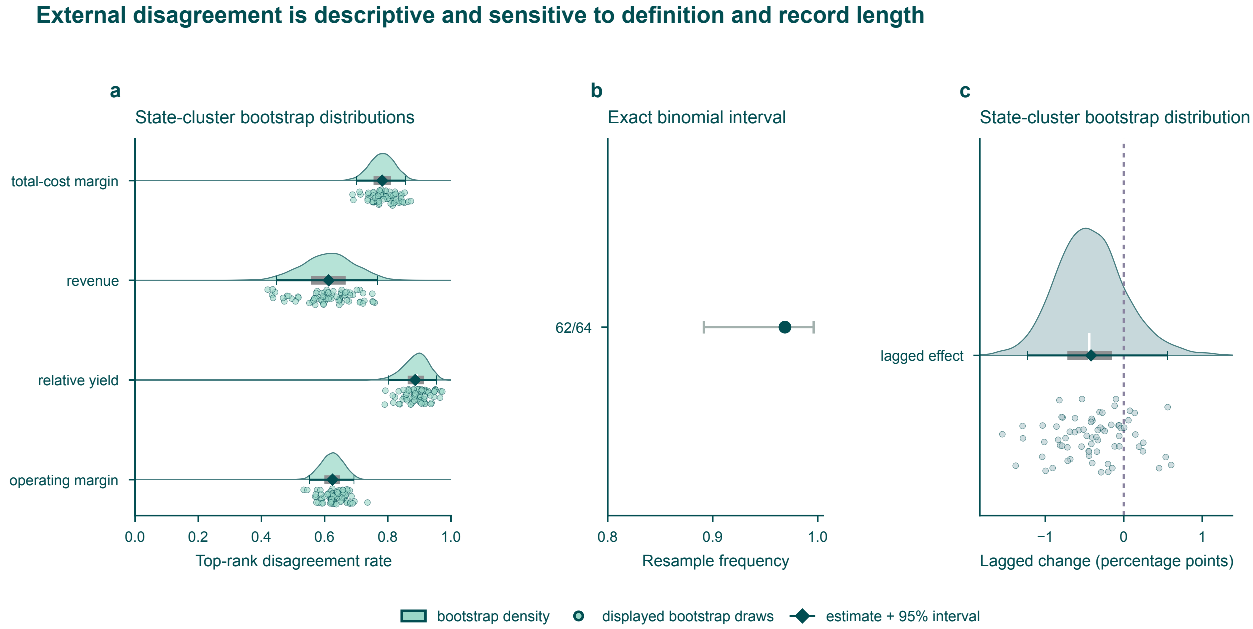

Figure 6: Resampling quantifies short-record sensitivity. **a**, Top-rank disagreement rates and state-cluster bootstrap distributions in 248 state-years. Half densities use all 5,000 draws, dots show 72 displayed bootstrap draws, thick bars show the interquartile range and white ticks the median; diamonds and thin bars are the observed estimates and bootstrap 95% intervals. **b**, Selected pairwise reversal occurs in 62 of 64 historical resamples; the bar is the exact Clopper–Pearson 95% binomial interval. **c**, Leakage-free lagged association between prior index leadership and next-year acreage-share change, shown with the same bootstrap distribution grammar as panel a.

interpreted as descriptive disagreement rather than a causal test of the mechanisms. These identification boundaries determine the managerial and policy implications developed next.

7 Managerial and policy implications

7.1 Recommendations should expose their decision layer

A crop-ranking tool can remain useful without prescribing acreage. The problem arises when an ordinal output is presented as if the top crop should receive the most land. A prescriptive system should expose at least five objects: the cardinal margin model, the joint distribution used to form portfolio losses, the operational feasible set, the risk convention and the selector used when optima are multiple. If these objects are unavailable, the defensible output is a rank or scenario comparison, not a claim of optimal acreage.

The distinction matters for accountability. A lender or platform that flags a farmer for planting less of a high-scoring crop may be penalizing compliance with a rotation, contract or liquidity restriction omitted from the score. Conversely, saying that every disagreement is rational would make the recommendation system unfalsifiable. The remedy is to compare declared feasible counterfactuals and characterize the optimal face.

7.2 Downside-risk governance

The Kansas experiment shows the cost of treating the selected Gaussian mean–variance policy as tail safe. Under matched rank dependence, that policy earns a higher mean but violates the Student-t evaluation-law loss-CVaR ceiling. This suggests a governance test for decision-support systems: report the allocation under at least one heavy-tailed joint stress law, evaluate it under the same downside-risk metric used for the farmer's tolerance, and disclose whether the policy remains feasible. The test is more informative than reporting a diversification index or pairwise correlation alone.

The result does not justify universal conservatism. A very tight risk ceiling reduces expected profit and can alter which crop dominates; a loose ceiling may leave the expected-profit allocation unchanged. Decision makers should therefore report the entire risk frontier and active-set transitions. The location of reversal is conditional on that frontier, not an immutable property of a crop.

7.3 Invest in information only with an action pathway

Forecast accuracy and operational flexibility should be evaluated jointly. When a share of acreage can be adjusted after a credible pre-plant signal, better information can be increasingly valuable. If planting is already fixed or posterior-optimal plans coincide, more signal accuracy has zero economic value even if the prediction is statistically informative. These are design criteria for climate services: document decision timing and the specific acreage, input, irrigation or harvest actions that remain open.

Shock-buffering flexibility creates a different investment trade-off. Irrigation, robust varieties, input switching or flexible harvest capacity can reduce exposure to the predicted state. This may raise total profit while reducing the incremental value of a forecast. Substitution is not a failure of the forecast; it means the operational technology has made the distinction between states less consequential. A portfolio of information and flexibility should therefore be assessed by total optimized value, not by requiring the interaction to be positive.

7.4 Policy interpretation

Rotation and diversification mandates should distinguish crop count from joint-tail protection. A policy that increases the number of planted crops can still leave a producer exposed to simultaneous drought losses when the crops share phenological vulnerabilities [Schmitt et al., 2024]. Crop-pair choice, timing, geography and access to off-farm risk instruments may matter more than a generic diversity threshold.

Likewise, a historical relative-yield performance index should not be evaluated solely against observed acreage. State acreage aggregates many farms and combines agronomic, financial and institutional decisions. The relevant policy question is whether the index improves a specified decision under realistic constraints. The external panel motivates that question but does not identify its causal answer. The conclusion synthesizes these conditional implications with the theoretical and numerical results.

8 Conclusion

Crop-ranking reversal occurs when an externally defined performance order and an optimized acreage order disagree. The analysis shows that this disagreement cannot be interpreted from the score alone. Expected margins, the joint distribution of losses, downside-risk tolerance and operational constraints jointly determine the allocation. In the Kansas calibration, winter wheat ranks first on the pre-decision historical relative-yield performance index but receives less land than corn and soybean. The resulting complete rank reversal reflects the decision model rather than a contradiction in the performance index.

The theoretical contribution is an exact two-crop reversal frontier under ordered expected margins. When the crop with the higher index also has the higher expected margin, reversal occurs if and only if the risk-feasible interval ends below the equal-share boundary. A unique dependence threshold requires continuity, monotonicity and crossing along a specified family path; without those conditions, reversal regions may be nonmonotone or disconnected. The complete KKT system identifies all marginal pressures, while optimal-face bounds distinguish selected, possible and universal pairwise reversal.

The three mechanism experiments separate economically different causes. The Kansas complete rank reversal is margin-induced because soybean has the higher expected margin and already dominates winter wheat at the expected-profit endpoint; the loss-CVaR ceiling moderates that ordering. A controlled mean-preserving downside shock produces risk-induced reversal by changing the soybean–corn allocation order while preserving both the index and expected-margin order. A separate rotation-cap path produces operationally induced reversal while margins and a nonbinding risk ceiling remain fixed. Positive lower bounds preclude strong exclusion reversal in the principal model. Allowing crop exit makes exclusion feasible, but the evaluated zero-lower-bound sensitivities still yield no strong exclusion reversal.

Diversification must likewise be evaluated against the risk measure governing the decision. The selected Gaussian mean–variance policy reduces variance relative to the expected-profit benchmark but has higher loss-CVaR than the tail-aware policy under the Student-t evaluation law and violates their common risk ceiling. Thus, a conventional variance improvement need not provide adequate protection against joint downside outcomes, even when the Gaussian and Student-t copulas share the same Kendall rank dependence. The comparison applies to the specified policy rule and sensitivity domain rather than to diversification in general.

Information and flexibility also require a conditional interpretation. Information value is nonnegative when the uninformed action remains feasible, and it is strictly positive under differing posterior optima in the separable or nonbinding-risk submodel. Flexibility can complement information when post-signal acreage adjustment expands useful responses. It can instead substitute for information when shock buffering contracts state-dependent margins toward a common vector. In the shared ex-ante loss-CVaR model, signed adjacent-grid cross-differences classify these interactions without extending the restricted theorem beyond its assumptions.

The 31-state panel documents the frequency of score–acreage disagreement under several ranking definitions, but it does not identify private objectives, constraints, beliefs or risk preferences. The Kansas margins also combine state yields with national prices and costs, the dependence parameters are structural stress values, and the eight-year calibration leaves crop shares and reversal boundaries imprecise. These boundaries limit causal and population interpretations while preserving the model-based mechanism results.

Future research should link pre-plant farm expectations, local prices and costs, crop histories, contracts, capacity and received signals to subsequent acreage decisions. Such data would allow the performance index, expected margins, feasible set and downside-risk tolerance to be estimated at a common decision level. The central conclusion is that an external crop ranking and an optimal acreage allocation answer different questions; their disagreement becomes scientifically interpretable only after the economic, risk and operational mechanisms are specified.

9 Methods

9.1 Data sources and transformations

State acreage and yield are obtained from USDA NASS annual reports [U.S. Department of Agriculture, National Agricultural Statistics Service, 2019, 2022, 2025]. National crop yield and harvest price are from USDA NASS, and operating and total costs are from USDA ERS [U.S. Department of Agriculture, Economic Research Service, 2026]. Annual CPI-U from the US Bureau of Labor Statistics converts monetary values to 2024 dollars [U.S. Bureau of Labor Statistics, 2026]. Crop-year panels are formed by complete-case joins; no missing or suppressed agricultural value is imputed.

The Kansas historical relative-yield performance index uses 2016–2023. For each crop and year, state yield is divided by the national yield of the same crop; the index is the eight-year mean. Margin is state yield times national harvest price minus national operating cost, then expressed in real 2024 dollars. Student-t, Gaussian and empirical-quantile marginals are calibrated to the historical means and dispersion. The primary index intentionally excludes the 2024 evaluation year.

9.2 Scenario generation and optimization

Common random numbers are used within each model contrast. Primary optimization uses 512 equally weighted scenarios and seed 202607270034. Copula correlation is derived from matched Kendall's τ. The Student-t copula uses four degrees of freedom and Student-t marginals use five. Linear programmes are solved with HiGHS. Every solution is re-evaluated from scenario profit, resource use and constraint violation, and its primal–dual conditions are checked.

Minimum loss-CVaR and expected-profit endpoints are solved separately for each joint law. Equation 8 produces eleven risk ceilings. To characterize objective-equivalent allocations, expected profit is constrained to its optimum within numerical tolerance and each ranked-pair difference is minimized and maximized. Complete rank reversal and exclusion feasibility are checked on the same optimal face. The deterministic HiGHS allocation defines the selected classification; universal and possible classifications use the face bounds. Ordering and zero tolerances are 10^{-4} normalized land share, with sensitivity from 10^{-8} to 10^{-2}.

The principal phase uses lower bounds 0.05, 0.10 and 0.05 for corn, soybean and winter wheat. Strong exclusion reversal is consequently infeasible by construction. Two complete sensitivity phases hold every other model component: one sets all three lower bounds to zero, and one sets only the highest-ranked winter wheat lower bound to zero. Each phase is re-solved at zero tolerances $10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}$ and 10^{-2}, including selected solutions and possible/universal classifications on the complete optimal face.

9.3 Diversification comparison

The benchmark x^0 maximizes expected profit under the matched Gaussian model. For each $\gamma \in \{0, 10^{-4}, \ldots, 0.03\}$, $x^{MV}(\gamma)$ maximizes Gaussian mean minus γ times Gaussian variance, subject to the same full-investment and operational constraints. The selected Gaussian mean–variance policy is chosen without reference to its tail outcome: it is the smallest γ that reduces Gaussian variance by at least 15% relative to x^0. Targets of 10% and 20% are sensitivity cases.

The tail-aware policy x^T maximizes expected profit subject to a common Student-t loss-CVaR ceiling. All three allocations are then evaluated under that Student-t law. Weak failure requires the fixed Gaussian variance target, a nonzero allocation difference and higher evaluation loss-CVaR for x^{MV} than for x^T. Strong failure additionally requires x^{MV} to violate the common ceiling. Sensitivity varies target, scenario count, seed, dependence strength, copula degrees of freedom, confidence level, risk tolerance and marginal family one factor at a time.

9.4 Mechanism isolation and robustness

For a raw heuristic x^h, the principal projection minimizes $\frac{1}{2}\|x - x^h\|_2^2$ subject to full investment, crop bounds, budget, rotation, soybean contract, planting-labour and harvest-equipment constraints. Idle land is not allowed in this comparison set. The loss-CVaR ceiling is excluded from projection and evaluated afterwards. The strictly convex problem is solved by SLSQP with objective tolerance 10^{-10}; feasibility is checked independently at 10^{-8}.

An alternative unweighted L^1 projection is solved by HiGHS. If its closest face is non-unique, a second linear programme minimizes a fixed crop-order weighted allocation with weights 1, 2 and 3 for corn, soybean and winter wheat, respectively, while remaining within 10^{-10} of the minimum L^1 distance. Raw targets, both projected allocations and both distances are reported. Expected-profit, full loss-CVaR and minimum-CVaR endpoints use the same constraints. Robustness varies one dimension at a time: confidence level, dependence family and parameter, marginal model, historical window, scenario count, seed, solver and removal of each constraint group. The analysis covers null, nonmonotone, multiple-optimum and infeasible cases.

The risk-isolation path preserves the Kansas soybean–corn index and mean orders, then applies a mean-preserving severe soybean downside event. Probability ranges from 2% to 20% and adverse gross-revenue share from 10% to 60%; risk tolerance spans 0 to 1. The operational path fixes margin scenarios and a nonbinding risk ceiling, adds the baseline calibrated constraint groups sequentially and then tightens only the soybean rotation cap. These controlled stresses identify mathematical mechanisms rather than estimating private farm constraints or shock probabilities.

Historical uncertainty resamples calibration years 64 times. Each replication re-estimates scores and margins, regenerates scenarios and resolves the frontier. Continuous quantities use historical-resample percentile intervals. Binary reversal frequencies use exact Clopper–Pearson binomial intervals. Neither interval treats simulated scenarios as independent empirical observations.

9.5 Information and flexibility

The binary state is constructed from the corn-minus-soybean relative-yield advantage around its historical median. Signal accuracy ranges from 0.5 to 1.0. A feasible constant policy is always available, providing an exact signal-ignorability check. The acreage-recourse path changes the share adjustable after signal arrival. The buffering path contracts state-dependent margins toward their common mean while preserving signal-contingent actions. One loss-CVaR ceiling applies to the full ex-ante state–signal loss distribution. Numerical information values below 10^{-7} are classified as zero. For each adjacent $\xi_1 < \xi_2$, $\phi_1 < \phi_2$ rectangle, interaction is the solved cross-difference $[V(\xi_2, \phi_2) - V(\xi_1, \phi_2)] - [V(\xi_2, \phi_1) - V(\xi_1, \phi_1)]$.

9.6 External descriptive analysis

For the 31-state panel, pairwise inversion intensity counts discordant pairs among the three crops. Top-rank disagreement equals one when the highest-index crop is not the largest-acreage crop. Confidence intervals use state-cluster resampling. The lagged model predicts next-year acreage-share change from the previous relative-yield leader and previous acreage share, with crop, state and decision-year fixed effects. It uses no contemporaneous index to predict the same year's acreage.

9.7 Data availability

The public source reports, analysis-ready tables and figure source data needed to reproduce the reported results will be deposited in a public archive upon publication. No confidential farm-level data are used.

9.8 Code availability

Analysis code sufficient to regenerate the numerical results and figures will be deposited with the source data upon publication. Additional mathematical proofs, phase classifications, data dictionaries and numerical diagnostics are provided in Supplementary Information.

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Author contributions

X.L. conducted the data analysis, developed the computational experiments and drafted the manuscript. P.C. conceived and supervised the study. Both authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

References

Farid AitSahlia, Chung-Jui Wang, Victor E. Cabrera, Stan Uryasev, and Clyde W. Fraisse. Optimal crop planting schedules and financial hedging strategies under ENSO-based climate forecasts. *Annals of Operations Research*, 190(1):203–217, 2011. doi: 10.1007/s10479-009-0551-2. URL `https://doi.org/10.1007/s10479-009-0551-2`.

Duc-Anh An-Vo, Ando Mariot Radanielson, Shahbaz Mushtaq, Kate Reardon-Smith, and Chris Hewitt. A framework for assessing the value of seasonal climate forecasting in key agricultural decisions. *Climate Services*, 22:100234, 2021. doi: 10.1016/j.cliser.2021.100234. URL `https://doi.org/10.1016/j.cliser.2021.100234`.

Jonathan Ansari and Marcus Rockel. Dependence properties of bivariate copula families. *Dependence Modeling*, 12(1), 2024. doi: 10.1515/demo-2024-0002. URL `https://doi.org/10.1515/demo-2024-0002`.

Mario Benini, Emanuele Blasi, Paolo Detti, and Lorenzo Fosci. Solving crop planning and rotation problems in a sustainable agriculture perspective. *Computers & Operations Research*, 159:106316, 2023. doi: 10.1016/j.cor.2023.106316. URL `https://doi.org/10.1016/j.cor.2023.106316`.

Dimitris Bertsimas and Nathan Kallus. From predictive to prescriptive analytics. *Management Science*, 66(3):1025–1044, 2020. doi: 10.1287/mnsc.2018.3253. URL `https://doi.org/10.1287/mnsc.2018.3253`.

Onur Boyabatlı, Javad Nasiry, and Yangfang Zhou. Crop planning in sustainable agriculture: Dynamic farmland allocation in the presence of crop rotation benefits. *Management Science*, 65(5):2060–2076, 2019. doi: 10.1287/mnsc.2018.3044. URL `https://doi.org/10.1287/mnsc.2018.3044`.

Stefano Demarta and Alexander J. McNeil. The t copula and related copulas. *International Statistical Review*, 73(1):111–129, 2005. doi: 10.1111/j.1751-5823.2005.tb00254.x. URL `https://doi.org/10.1111/j.1751-5823.2005.tb00254.x`.

Xiaodong Du, David A. Hennessy, Hongli Feng, and Gaurav Arora. Land resilience and tail dependence among crop yield distributions. *American Journal of Agricultural Economics*, 100(3):809–828, 2018. doi: 10.1093/ajae/aax082. URL `https://doi.org/10.1093/ajae/aax082`.

Adam N. Elmachtoub and Paul Grigas. Smart predict, then optimize. *Management Science*, 68(1):9–26, 2022. doi: 10.1287/mnsc.2020.3922. URL `https://doi.org/10.1287/mnsc.2020.3922`.

Carlo Filippi, Renata Mansini, and Elisa Stevanato. Mixed integer linear programming models for optimal crop selection. *Computers & Operations Research*, 81:26–39, 2017. doi: 10.1016/j.cor.2016.12.004. URL `https://doi.org/10.1016/j.cor.2016.12.004`.

Jeffrey M. Keisler. Value of information in portfolio decision analysis. *Decision Analysis*, 1(3):177–189, 2004. doi: 10.1287/deca.1040.0023. URL `https://doi.org/10.1287/deca.1040.0023`.

Ryan Larsen, David J. Leatham, and Kunlapath Sukcharoen. Geographical diversification in wheat farming: A copula-based CVaR framework. *Agricultural Finance Review*, 75(3):368–384, 2015. doi: 10.1108/AFR-07-2014-0020. URL `https://doi.org/10.1108/AFR-07-2014-0020`.

Jing Li, Daniel Rodriguez, Dongqing Zhang, and Kaiping Ma. Crop rotation model for contract farming with constraints on similar profits. *Computers and Electronics in Agriculture*, 119:12–18, 2015. doi: 10.1016/j.compag.2015.10.002. URL `https://doi.org/10.1016/j.compag.2015.10.002`.

Konstantinos G. Liakos, Patrizia Busato, Dimitrios Moshou, Simon Pearson, and Dionysis Bochtis. Machine learning in agriculture: A review. *Sensors*, 18(8):2674, 2018. doi: 10.3390/s18082674. URL `https://doi.org/10.3390/s18082674`.

Jayanta Mandi, James Kotary, Senne Berden, Maxime Mulamba, Victor Bucarey, Tias Guns, and Ferdinando Fioretto. Decision-focused learning: Foundations, state of the art, benchmark and future opportunities. Technical Report 2307.13565, arXiv, 2023. URL `https://arxiv.org/abs/2307.13565`.

Miley Wesson Merkhofer. The value of information given decision flexibility. Technical Report ADA016836, Defense Technical Information Center, 1975. URL `https://doi.org/10.21236/ADA016836`.

Francisco J. Meza, James W. Hansen, and Daniel Osgood. Economic value of seasonal climate forecasts for agriculture: Review of ex-ante assessments and recommendations for future research. *Journal of Applied Meteorology and Climatology*, 47(5):1269–1286, 2008. doi: 10.1175/2007JAMC1540.1. URL `https://doi.org/10.1175/2007JAMC1540.1`.

Thong Nguyen-Huy, Ravinesh C. Deo, Shahbaz Mushtaq, Jarrod Kath, and Shahjahan Khan. Copula-based agricultural conditional value-at-risk modelling for geographical diversifications in wheat farming portfolio management. *Weather and Climate Extremes*, 21:76–89, 2018. doi: 10.1016/j.wace.2018.07.002. URL https://doi.org/10.1016/j.wace.2018.07.002.

Christoph Pahmeyer, Thomas Kuhn, and Wolfgang Britz. Fruchtfolge: A crop rotation decision support system for optimizing cropping choices with big data and spatially explicit modeling. *Computers and Electronics in Agriculture*, 181:105948, 2021. doi: 10.1016/j.compag.2020.105948. URL https://doi.org/10.1016/j.compag.2020.105948.

Georg Pflug, Franziska Gaupp, Stefan Hochrainer-Stigler, Jim Hall, and Simon Dadson. Dependency of crop production between global breadbaskets. *Risk Analysis*, 37(11):2212–2228, 2017. doi: 10.1111/risa.12761. URL https://doi.org/10.1111/risa.12761.

Marcus Randall, James Montgomery, and Andrew Lewis. Robust temporal optimisation for a crop planning problem under climate change uncertainty. *Operations Research Perspectives*, 9:100219, 2022. doi: 10.1016/j.orp.2021.100219. URL https://doi.org/10.1016/j.orp.2021.100219.

Marcus Randall, Karin Schiller, Andrew Lewis, James Montgomery, and Muhammad Shahinur Alam. A systematic review of crop planning optimisation under climate change. *Water Resources Management*, 38(6):1867–1881, 2024. doi: 10.1007/s11269-024-03758-3. URL https://doi.org/10.1007/s11269-024-03758-3.

R. Tyrrell Rockafellar and Stanislav Uryasev. Optimization of conditional value-at-risk. *The Journal of Risk*, 2(3):21–41, 2000. doi: 10.21314/JOR.2000.038. URL https://doi.org/10.21314/JOR.2000.038.

R. Tyrrell Rockafellar and Stanislav Uryasev. Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7):1443–1471, 2002. doi: 10.1016/S0378-4266(02)00271-6. URL https://doi.org/10.1016/S0378-4266(02)00271-6.

Jonas Schmitt, Frank Offermann, Andreia F. S. Ribeiro, and Robert Finger. Drought risk management in agriculture: A copula perspective on crop diversification. *Agricultural Economics*, 55(5):823–847, 2024. doi: 10.1111/agec.12851. URL https://doi.org/10.1111/agec.12851.

U.S. Bureau of Labor Statistics. Consumer price index: All urban consumers, all items in u.s. city average, series cuur0000sa0. CPI database, 2026. URL https://www.bls.gov/cpi/data.htm. Monthly observations through 2024; accessed 27 July 2026.

U.S. Department of Agriculture, Economic Research Service. Commodity costs and returns. Data product, 2026. URL https://www.ers.usda.gov/data-products/commodity-costs-and-returns/. Frozen recent-series crop files dated 1 May 2026; accessed 27 July 2026.

U.S. Department of Agriculture, National Agricultural Statistics Service. Crop production 2018 summary. Technical report, U.S. Department of Agriculture, January 2019. URL https://www.nass.usda.gov/Publications/Todays_Reports/reports/cropan19.pdf.

U.S. Department of Agriculture, National Agricultural Statistics Service. Crop production 2021 summary. Technical report, U.S. Department of Agriculture, January 2022. URL https://www.nass.usda.gov/Publications/Todays_Reports/reports/cropan22.pdf.

U.S. Department of Agriculture, National Agricultural Statistics Service. Crop production 2024 summary. Technical report, U.S. Department of Agriculture, January 2025. URL https://www.nass.usda.gov/Publications/Todays_Reports/reports/cropan25.pdf.

Thomas van Klompenburg, Ayalew Kassahun, and Cagatay Catal. Crop yield prediction using machine learning: A systematic literature review. *Computers and Electronics in Agriculture*, 177:105709, 2020. doi: 10.1016/j.compag.2020.105709. URL https://doi.org/10.1016/j.compag.2020.105709.

Jan A. Van Mieghem. Investment strategies for flexible resources. *Management Science*, 44(8):1071–1078, 1998. doi: 10.1287/mnsc.44.8.1071. URL https://doi.org/10.1287/mnsc.44.8.1071.